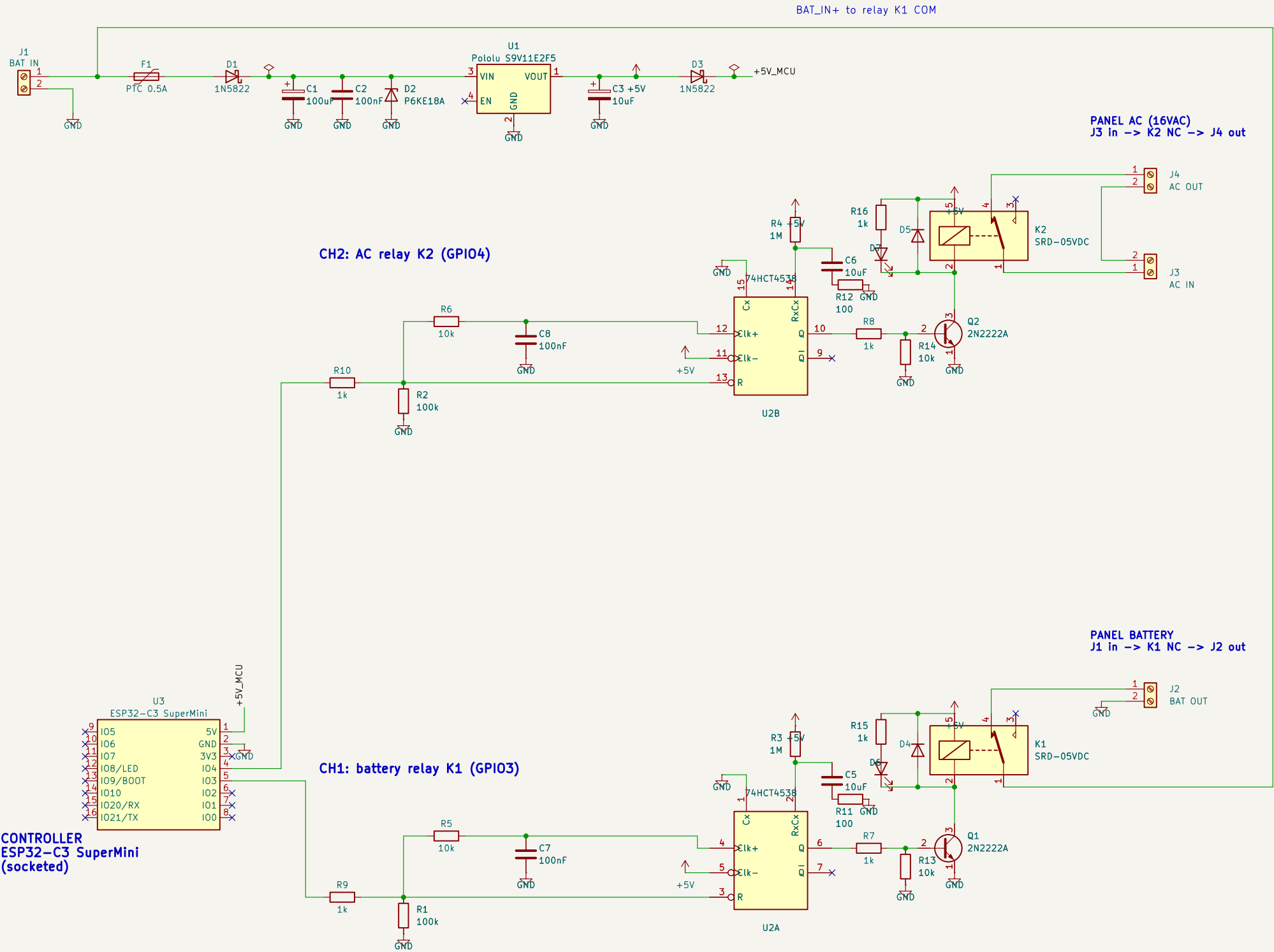
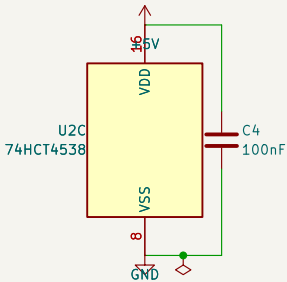


POWER: J1 battery tap (never switched) -> F1 PTC -> D1 reverse-polarity -> U1 Pololu 5V. D3 blocks USB backfeed from the SuperMini onto the +5V rail



HARDWARE TIMEOUT (74HCT4538 @ 5V)
Relay ON = GPIO high AND < T since rising edge
 $T = 0.7 \times R_{ext} \times C_{ext} = 0.7 \times 1M \times 10uF = \sim 7s$
(Tasmota PulseTime 3.0 / 3.5 s is always shorter)
R9/R10 1k: protect ESP GPIO if 5V rail is off (USB only)
R11/R12 100R: TI rapid power-down protection for Cext

RELAY DRIVERS
Q -> R7/R8 1k -> 2N2222A (R13/R14 10k base pull-down)
D4/D5 1N4007 flyback across each coil
R15/R16 1k + D6/D7 red LED: lit while the relay is energised
Fail-safe: Q low / U2 unpowered -> coil off -> NC closed



U2C = power pins of U2 (same 74HCT4538 chip as U2A/U2B)
Pin 16 VDD = +5V, pin 8 VSS = GND
C4 100nF decoupling: place right beside pins 16 and 8 on the PCB

On-board SRD-05VDC-SL-C relays, NC contacts in panel power paths
ESP32-C3 (Tasmota) + 74HCT4538 hardware timeout

Sheet: /
File: texecom-power-reset.kicad_sch
Title: Texecom Remote Power Reset – carrier PCB

Size: A3 Date: 2026-09-25 Rev: A
KiCad E.D.A. 10.0.5 Id: 1/1